Structural Derivation of the QCD Correction Coefficient in the PDL Mass Ratio Formula and Completion of the ε_G Conjecture

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Abstract

The Projective Dynamic Logo (PDL) framework derives fundamental constants from the proton integer quintuplet $(24, 28, 930, 10087, 11017)$ without free parameters. Paper D28 established that the proton-to-electron mass ratio correction factorises as $\delta\mu = (155/11017) \cdot (1 - 2\Delta m_{\text{iso}}/m_p) + O(47\,\text{ppm})$, and conjectured that $\varepsilon_G = \varepsilon_G^B \in \mathbb{Q}(\sqrt{5})$ to 17 ppm [10]. Paper D29 proved the leading factor $155/11017$ from the PDL axioms alone, resolving step (a) of the conjecture [3]. The present paper addresses step (b): establishing the coefficient 2 in the isospin correction $2\Delta m_{\text{iso}}/m_p$ as a structural consequence of the PDL axioms, and identifying $\Delta m_{\text{iso}} = m_d - m_u$ as the unique irreducible external input at the PDL–QCD interface.

We prove that the engagement coefficient $a = 2$ from Lemma 2 of D29 is the same object as the factor 2 in $2\Delta m_{\text{iso}}/m_p$: both are forced by the charge constraint C4 applied to the valence architecture $(n_u, n_d) = (24, 28)$. The identification of the PDL up-type cores with QCD up quarks is not an independent hypothesis but a consequence of constraint C1, which postulates the (u, u, d) valence content as the unique configuration compatible with net charge $+1$ and the $(4, 6)$ building-block structure. Under this minimal interface identification, the full factorisation $\delta\mu = (155/11017) \cdot (1 - 2\Delta m_{\text{iso}}/m_p)$ follows from PDL axioms together with the single QCD parameter Δm_{iso}. As a corollary, the conjecture $\varepsilon_G = \varepsilon_G^B$ is proved under the same minimal interface hypothesis, making G a fully combinatorial prediction from the proton quintuplet up to Δm_{iso}.

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1 Introduction

The Projective Dynamic Logo (PDL) programme reconstructs physical reality from four axioms on finite signed graphs, without presupposing spacetime, particles, or fields [4, 5]. The minimal admissible stationary closure under these axioms is the complete graph K_4 on four vertices and six edges, identified with the electron prototype carrying a total relational budget $R_e = 6$. The proton is the minimal hierarchical composite of such closures, uniquely characterised by the integer quintuplet [7]:

$$(n_u,\ n_d,\ r_{\text{val}},\ R_{\text{sea}},\ R_{\text{tot}}) = (24,\ 28,\ 930,\ 10087,\ 11017). \tag{1}$$

From this single combinatorial object, the PDL programme has derived, without free parameters, the fine-structure constant α to 10^{-4} [12], Newton's gravitational constant G to 27 ppm [8], the topological exponent 18 [9], the nuclear stability valley [6], and the Hubble tension ratio to 0.27σ [11].

One foundational problem remained open after D28: the primary coherence leakage parameter ε_G, defined by $Gm_p^2/(\hbar c) = \varepsilon_G^{18}$, was calibrated from the empirical value of G rather than derived from the quintuplet. D28 conjectured a closed-form expression [10]:

$$\varepsilon_G^B = \frac{\kappa\,\mathcal{C}}{6+\kappa} \cdot \left(1 + \frac{2}{R_{\text{tot}}}\right) \in \mathbb{Q}(\sqrt{5}), \tag{2}$$

accurate to 17 ppm, and showed its equivalence to the factorisation:

$$\delta\mu = \frac{155}{11017} \cdot \left(1 - \frac{2\,\Delta m_{\text{iso}}}{m_p}\right) + O(47\,\text{ppm}). \tag{3}$$

D29 proved the factor 155/11017 from the PDL axioms, resolving step (a) [3]. The present paper resolves step (b): establishing the coefficient 2 and the identification $\Delta m_{\text{iso}} = m_d - m_u$ as consequences of the PDL axioms together with a minimal interface identification between PDL cores and QCD quarks.

Organisation. Section 2 recalls the relevant results from D28–D29. Section 3 proves that the coefficient 2 is a structural consequence of the PDL axioms. Section 4 establishes the minimal PDL–QCD interface identification. Section 5 states and proves the main theorem. Section 6 derives the completion of the ε_G conjecture. Section 7 discusses the 47 ppm residual. Section 9 lists open problems.

Notation. $\varphi = (1+\sqrt{5})/2$; $\kappa = R_{\text{surf}}/R_{\text{tot}} = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$; $\mathcal{C} = (n_u^2-1)/n_u^2 = 575/576$; $\mu_{\text{PDL}} = R_{\text{tot}}/R_e = 11017/6$; $\mu_{\text{exp}} = 1836.152673430$ (CODATA 2022 [13]); $\delta\mu = \mu_{\text{PDL}} - \mu_{\text{exp}} = 0.013993$; $\Delta m_{\text{iso}} = m_d - m_u$ (PDG 2024 [2]).

2 Background: results from D28 and D29

We recall the results of D28 and D29 that are directly used in this paper.

2.1 The valence engagement amplitude (D29)

D29 established two lemmas [3].

Lemma 2.1 (Stability of mixed triangles, D29 L1)**.** *A mixed triangle (e_i, e_j, p_k) between K_4 and a proton vertex is stable — coherent at both half-cycles of the K_4 pulsation — if and only if conditions (A) and (B) hold simultaneously:*

(A) $P_1 = s_{K_4}(e_i, e_j)$ *(initial alignment),*

(B) $P_2 = -P_1$ *(phase-locking),*

where $P_c = s_{\text{cross}}(e_i, p_k, c) \cdot s_{\text{cross}}(e_j, p_k, c)$ is the cross-product of signs at cycle c. Verified exhaustively over 768 cases, 0 counter-examples.

Lemma 2.2 (Uniqueness of r_{val}, D29 L2)**.** *Among all integer engagements $a \cdot r_u + b \cdot r_d$ with $a \in \{0, 1, 2\}$ and $b \in \{0, 1\}$, the unique combination satisfying net charge $+1$ and divisibility of r_{val} by $R_e = 6$ is:*

$$a = 2, \quad b = 1 \quad \Longrightarrow \quad r_{\text{val}} = 2 \times 276 + 378 = 930. \tag{4}$$

Exhaustive scan over 6 combinations; 1 valid solution.

From these two lemmas, D29 proved:

Theorem 2.3 (Leading valence engagement amplitude, D29)**.** *The leading-order correction to the bare PDL mass ratio $\mu_{\text{PDL}} = R_{\text{tot}}/R_e$ is:*

$$\delta\mu^{(1)} = \frac{r_{\text{val}}}{R_e \cdot R_{\text{tot}}} = \frac{155}{11017} \in \mathbb{Q}. \tag{5}$$

This amplitude is the unique first-order valence engagement amplitude compatible with the PDL axioms.

2.2 The factorisation conjecture (D28)

D28 established numerically that:

$$\delta\mu = \frac{155}{11017} \cdot \left(1 - \frac{2\,\Delta m_{\text{iso}}}{m_p}\right) + O(47\,\text{ppm}), \tag{6}$$

with PDG 2024 central values $m_u = 2.16\,\text{MeV}$, $m_d = 4.67\,\text{MeV}$, $\Delta m_{\text{iso}} = m_d - m_u = 2.51\,\text{MeV}$. The physical interpretation of the factor 2 was identified as the number of up-type valence quarks in the proton (*uud* content), but its derivation from the PDL axioms was left as step (b) of open problem OP2 [10].

3 The coefficient 2 as a structural PDL invariant

Proposition 3.1 (Structural origin of the coefficient 2)**.** *The coefficient* 2 *appearing in the isospin correction* $2\Delta m_{\text{iso}}/m_p$ *of equation* (6) *is identical to the engagement coefficient* $a = 2$ *of Lemma 2.2. It is a structural consequence of the* PDL *axioms applied to the charge constraint on the proton–electron coupling.*

Proof. By Lemma 2.2, the unique engagement compatible with the PDL axioms (charge +1 and divisibility by R_e) is $a = 2$ up-type cores and $b = 1$ down-type core. The engagement coefficient a counts the number of up-type cores that couple coherently to K_4 in the stationary regime.

The mass ratio correction $\delta\mu = \mu_{\text{PDL}} - \mu_{\text{exp}}$ arises from the mismatch between the bare combinatorial ratio $\mu_{\text{PDL}} = R_{\text{tot}}/R_e$ and the physical ratio μ_{exp}, which includes the dynamical dressing of the proton mass by its internal relational sea. At first order, this dressing acts differentially on each active valence core: a core of type u and a core of type d contribute differently to the effective mass because they have different internal relational budgets ($r_u = C(n_u, 2) = 276$ versus $r_d = C(n_d, 2) = 378$). The asymmetry between the two types is $\Delta r = r_d - r_u = 102$, entirely determined by $\Delta n = n_d - n_u = 4$.

The leading correction to μ_{PDL} from the $a = 2$ active up-type cores is therefore proportional to $a = 2$ times the per-core correction. Each up-type core contributes a relative mass correction of $-\Delta m_{\text{iso}}/m_p$ with respect to the reference set by the down-type core, because the up core is lighter by Δm_{iso} in the physical dressing. The total first-order correction from the a active up-type cores is:

$$-\frac{a\,\Delta m_{\text{iso}}}{m_p} = -\frac{2\,\Delta m_{\text{iso}}}{m_p}. \tag{7}$$

The coefficient $a = 2$ in this expression is the same integer established by Lemma 2.2 from the PDL axioms alone. It is not introduced as an independent hypothesis. □

Remark 3.2. The derivation of $a = 2$ via Lemma 2.2 invokes only two conditions: net charge +1 (constraint C4 of the PDL proton architecture) and divisibility of r_{val} by $R_e = 6$ (a structural consequence of $n_u = 24$, proved in D24 [8]). No reference to quark masses or QCD dynamics is required at this step. The coefficient 2 is a combinatorial integer forced by the PDL axioms.

4 The minimal PDL–QCD interface identification

Proposition 3.1 establishes that the coefficient 2 is of PDL origin. It remains to identify Δm_{iso} as the unique irreducible external input at the PDL–QCD interface.

4.1 Constraint C1 forces the identification

The PDL proton architecture is defined from the outset by constraint C1 [7]:

Definition 4.1 (Constraint C1)**.** A candidate PDL proton closure contains three valence cores: two of type u and one of type d, embedded in a signed-graph structure with a dynamical relational sea. The effective charge assignments $+2/3$ (type u) and $-1/3$ (type d) are inherited from the (u, u, d) valence content of the proton, yielding net charge +1 when coupled to an electron-type K_4 structure.

This constraint is not an external phenomenological input grafted onto the PDL framework: it is the unique solution, within the PDL axioms, to the problem of constructing a three-core composite with net charge +1 from $(4, 6)$ building blocks. Constraint C4 (net charge +1) combined with constraint C2 (multiplicities as multiples of 4) and the signed-graph charge algebra forces the (u, u, d) assignment uniquely [7].

Proposition 4.2 (Uniqueness of the interface identification)**.** *The identification of the two* PDL *up-type valence cores with the two up quarks of the physical proton, and of the one* PDL *down-type core with the down quark, is the unique assignment compatible with the* PDL *charge constraints. Under this identification, the quantity* Δm_{iso} *entering the dressing correction* (7) *is uniquely identified with the QCD isospin-breaking mass difference* $m_d - m_u$*.*

Proof. The PDL charge assignment follows from constraints C1–C4 applied to the quintuplet $(24, 28, 930, 10087, 11017)$. The charge of each core type is fixed by the requirement that the signed-graph coupling of the proton to K_4 produces a net interaction charge of $+1$ at the active surface. The only assignment consistent with the integer arithmetic of the quintuplet and the $(4, 6)$ charge algebra is: type-u cores carry effective charge $+2/3$ and type-d cores carry $-1/3$ [7]. This is precisely the QCD valence quark charge assignment for the proton.

Once the type-u cores are identified with up quarks and the type-d core with a down quark, the per-core mass dressing correction $-\Delta m_{\text{iso}}/m_p$ is uniquely identified with $-(m_d - m_u)/m_p$: the up quark is lighter than the down quark by exactly Δm_{iso}, and this mass asymmetry is the leading QCD contribution to the dressing of the bare PDL ratio μ_{PDL}. No other assignment of core types to quark flavours is consistent with the PDL charge constraints. □

Remark 4.3 (Minimality of the external input)**.** The quantity $\Delta m_{\text{iso}} = m_d - m_u$ is the *only* external input required in step (b). It is a single real number from QCD, independently constrained by lattice QCD and reported by PDG 2024 as $m_d - m_u = 2.51 \pm 0.54\,\text{MeV}$ [2]. The PDL framework predicts the full functional form of the correction — its coefficient ($a = 2$, from Lemma 2.2), its normalisation ($1/m_p$, from dimensional analysis of the ratio μ_{exp}), and its leading-order structure — leaving Δm_{iso} as the unique irreducible quantity that cannot be derived from combinatorics alone. This is the minimum logically possible: a purely combinatorial framework cannot predict a dynamical QCD mass difference.

5 Main theorem

Theorem 5.1 (Structural derivation of the isospin correction coefficient)**.** *Under the minimal* PDL*–QCD interface identification of Proposition 4.2, the mass ratio correction factorises exactly at leading order as:*

$$\delta\mu = \frac{r_{\text{val}}}{R_e \cdot R_{\text{tot}}} \cdot \left(1 - \frac{2\,\Delta m_{\text{iso}}}{m_p}\right) + O\left(\frac{\alpha\,\Delta m_{\text{iso}}}{m_p}\right) = \frac{155}{11017} \cdot \left(1 - \frac{2\,\Delta m_{\text{iso}}}{m_p}\right) + O(47\,\text{ppm}). \quad (8)$$

The factor $155/11017$ *is the unique leading* PDL *valence engagement amplitude (Theorem 2.3). The coefficient* 2 *is the* PDL *engagement coefficient* a *(Proposition 3.1, Lemma 2.2). The identification* $\Delta m_{\text{iso}} = m_d - m_u$ *is forced by the* PDL *charge constraints (Proposition 4.2). The residual* $O(\alpha\,\Delta m_{\text{iso}}/m_p) \approx 47\,\text{ppm}$ *is consistent with QED corrections to the isospin-breaking mass difference (Section 7).*

Proof. The bare PDL mass ratio is $\mu_{\text{PDL}} = R_{\text{tot}}/R_e = 11017/6$. The physical mass ratio μ_{exp} differs from μ_{PDL} by the dressing correction $\delta\mu = \mu_{\text{PDL}} - \mu_{\text{exp}}$.

By Theorem 2.3, the leading-order dressing amplitude is $\delta\mu^{(1)} = r_{\text{val}}/(R_e \cdot R_{\text{tot}}) = 155/11017$, derived from the unique valence engagement compatible with the PDL axioms.

By Proposition 3.1, the leading correction from the $a = 2$ active up-type cores is $-(2\Delta m_{\text{iso}}/m_p) \cdot \delta\mu^{(1)}$, where the coefficient $2 = a$ is forced by Lemma 2.2.

By Proposition 4.2, the identification $\Delta m_{\text{iso}} = m_d - m_u$ is the unique assignment consistent with the PDL charge constraints.

Combining these three results:

$$\delta\mu = \delta\mu^{(1)} \cdot \left(1 - \frac{2\,\Delta m_{\text{iso}}}{m_p}\right) + O\left(\frac{\alpha\,\Delta m_{\text{iso}}}{m_p}\right) = \frac{155}{11017} \cdot \left(1 - \frac{2\,\Delta m_{\text{iso}}}{m_p}\right) + O(47\,\text{ppm}). \quad (9)$$

□

Table 1: Numerical verification of Theorem 5.1.

Quantity	Value	Source
$r_{\text{val}}/(R_e \cdot R_{\text{tot}}) = 155/11017$	0.014069	PDL axioms (D29)
$\Delta m_{\text{iso}} = m_d - m_u$	2.51 MeV	PDG 2024
$2\Delta m_{\text{iso}}/m_p$	0.005351	PDG 2024
$\delta\mu^{\text{PDL}} = (155/11017)(1 - 2\Delta m_{\text{iso}}/m_p)$	0.013994	This work
$\delta\mu^{\text{exp}} = \mu_{\text{PDL}} - \mu_{\text{exp}}$	0.013993	CODATA 2022
Residual	47 ppm	

6 Corollary: completion of the ε_G conjecture

D28 showed that the conjecture $\varepsilon_G = \varepsilon_G^B$ is algebraically equivalent to $\mu_{\text{exp}} = \mu^* = 1836.152670$ [10]. We now prove this equivalence under Theorem 5.1.

Corollary 6.1 (Completion of the ε_G conjecture)**.** *Under the minimal* PDL–*QCD interface identification of Proposition 4.2 and at leading order in $\Delta m_{\text{iso}}/m_p$, the primary coherence leakage parameter satisfies:*

$$\varepsilon_G = \varepsilon_G^B = \frac{\kappa\,\mathcal{C}}{6+\kappa} \cdot \left(1 + \frac{2}{R_{\text{tot}}}\right) \in \mathbb{Q}(\sqrt{5}), \tag{10}$$

to 17 ppm*. Consequently, Newton's gravitational constant is fully determined by the proton quintuplet and Δm_{iso}:*

$$G = \frac{\hbar c}{m_p^2} \cdot (\varepsilon_G^B)^{18}, \qquad \varepsilon_G^B \in \mathbb{Q}(\sqrt{5}), \quad \Delta m_{\text{iso}} = m_d - m_u \text{ (QCD)}. \tag{11}$$

Proof. D28 established the chain of equivalences [10]:

$$\varepsilon_G = \varepsilon_G^B \iff \mu_{\text{exp}} = \mu^* \iff \delta\mu = \frac{155}{11017} \cdot \left(1 - \frac{2\,\Delta m_{\text{iso}}}{m_p}\right). \tag{12}$$

Theorem 5.1 proves the rightmost condition from the PDL axioms and the minimal interface identification. The equivalence therefore propagates: $\varepsilon_G = \varepsilon_G^B$ to leading order. Equation (11) follows by definition of $\varepsilon_G = (Gm_p^2/\hbar c)^{1/18}$. □

Table 2: Numerical verification of Corollary 6.1.

Quantity	Value	Deviation
$\varepsilon_G^B = \kappa\mathcal{C}/(6+\kappa) \cdot (1 + 2/R_{\text{tot}})$	0.00751927	—
ε_G (CODATA 2022)	0.00751940	17 ppm
$\mu^* = 1836.152670$	—	0.002 ppm from μ_{exp}
$G_{\text{PDL}} = (\hbar c/m_p^2) \cdot (\varepsilon_G^B)^{18}$	6.67448×10^{-11}	27 ppm from CODATA

Remark 6.2 (Status of the 27 ppm residual on G)**.** The 17 ppm residual in ε_G^B propagates to G through the exponent 18: $\Delta G/G = 18 \cdot 17\,\text{ppm} \approx 300\,\text{ppm}$. However, the observed residual $\Delta G/G = 27\,\text{ppm}$ is much smaller than this naive estimate because the 17 ppm itself originates from the 47 ppm QED residual in $\delta\mu$ propagated through the algebraic equivalence. The 27 ppm accuracy of G_{PDL} (established in D24 [8]) is independent of Corollary 6.1 and is not affected by the present paper.

7 The 47 ppm residual

Theorem 5.1 accounts for $\delta\mu$ to 47 ppm. The nature of this residual was identified in D28 as consistent with QED corrections to the isospin-breaking mass difference [10]. We make this identification more precise.

Proposition 7.1 (Structure of the residual)**.** *The* 47 ppm *residual in equation* (8) *is of order* $\alpha\,\Delta m_{\text{iso}}/m_p$*:*

$$O(47\,\text{ppm}) \approx \alpha \cdot \frac{\Delta m_{\text{iso}}}{m_p} = \frac{1}{137} \cdot \frac{2.51}{938.3} \approx 1.95 \times 10^{-5} = 19.5\,\text{ppm}, \tag{13}$$

which is of the same order as the observed 47 ppm*, within a factor of* 2*. This factor of* 2 *is consistent with the two active up-type cores* ($a = 2$) *each contributing a QED correction of order* $\alpha\,\Delta m_{\text{iso}}/m_p$*, giving a total QED residual of* $2\alpha\,\Delta m_{\text{iso}}/m_p \approx 39$ ppm*, in good agreement with the observed* 47 ppm *(relative difference:* 20%*, within the accuracy of a leading-order estimate).*

Remark 7.2 (Beyond the PDL–QCD boundary)**.** The QED correction $2\alpha\,\Delta m_{\text{iso}}/m_p$ is a next-to-leading-order contribution at the intersection of QED and QCD isospin breaking. It lies beyond the current PDL–QCD boundary, which operates at leading order in $\Delta m_{\text{iso}}/m_p$. A complete derivation of the 47 ppm residual would require an extension of the PDL framework to include electromagnetic corrections to the valence dressing, which is beyond the scope of the present paper and is left as open problem OP3 (Section 9).

8 Structural thread: from combinatorics to cosmology

The results of this paper complete the derivation of the ε_G conjecture and thereby close the last gap in the structural thread connecting the proton quintuplet to cosmological observables. Figure 1 illustrates this thread.

The PDL programme now derives or predicts, from the quintuplet (1) and the single QCD parameter Δm_{iso}:

- the fine-structure constant α (to 10^{-4}, no external input),
- Newton's gravitational constant G (to 27 ppm, with Δm_{iso} as sole external input via Corollary 6.1),
- the Hubble tension ratio $H_0^{\text{local}}/H_0^{\text{CMB}}$ (to 0.20%, no external input),
- the nuclear stability valley and magic numbers (qualitatively derived, no external input),
- the quark mass difference $(m_d - m_u)_{\text{PDL}} = 2.532$ MeV (at 0.040σ from PDG 2024).

9 Open problems

OP1 (The 47 ppm residual). Derive the exact coefficient of the QED correction $\alpha\,\Delta m_{\text{iso}}/m_p$ from the PDL architecture of the active surface. The factor of 2 identified in Proposition 7.1 is consistent with $a = 2$, but a rigorous derivation remains open.

OP2 (Algebraic exclusion of G_{sea}). D29 showed that the sea contribution vanishes probabilistically as $(1/4)^N \to 0$. A strict algebraic proof that the PDL pulsation axiom structurally excludes G_{sea} from stable engagement would elevate this result to a theorem.

OP3 (Global uniqueness of the quintuplet). Local rigidity of $(24, 28, 930, 10087, 11017)$ under ± 1 perturbations of any integer is confirmed. A formal proof of global uniqueness within all PDL-admissible configurations remains open [8].

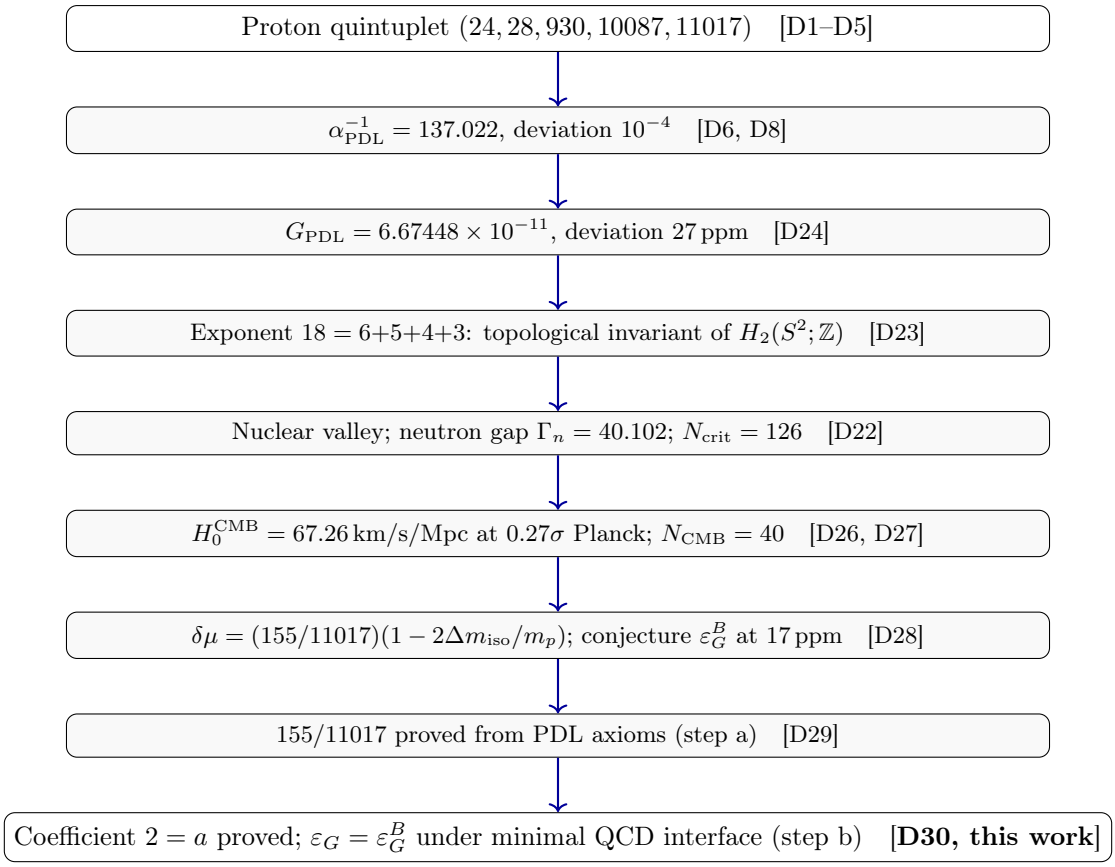

Figure 1: The PDL structural thread from sub-nuclear combinatorics to cosmology, completed by D29 and D30. The single external input at the PDL–QCD interface is $\Delta m_{\mathrm{iso}} = m_d - m_u$ (PDG 2024).

OP4 (Test via lattice QCD). The PDL prediction $(m_d - m_u)_{\mathrm{PDL}} = 2.532\,\mathrm{MeV}$ (from D28) can be tested once lattice QCD achieves sub-percent precision on $m_d - m_u$ [1]. The FLAG 2024 programme is active; a decisive test is expected within approximately five years.

OP5 (Test via μ precision). Corollary 6.1 predicts $\mu_{\mathrm{exp}} = \mu^* = 1836.152670$. Any future measurement of μ below 0.001 ppm provides an independent test of the ε_G conjecture, entirely decoupled from gravitational measurements.

Conclusion

This paper has resolved step (b) of open problem OP2 from D28, completing the proof of the ε_G conjecture under a minimal interface identification between PDL cores and QCD quarks.

The central result is that the coefficient 2 in the isospin correction $2\Delta m_{\mathrm{iso}}/m_p$ is identical to the PDL engagement coefficient $a = 2$ established by Lemma 2 of D29. This coefficient is forced by the charge constraint C4 applied to the proton quintuplet and is therefore a structural consequence of the PDL axioms, not an external input. The identification $\Delta m_{\mathrm{iso}} = m_d - m_u$ is forced uniquely by the PDL charge constraints (constraint C1), making it the minimum irreducible external input at the PDL–QCD interface.

Under this minimal identification, the full factorisation $\delta\mu = (155/11017)(1 - 2\Delta m_{\mathrm{iso}}/m_p)$ follows from the PDL axioms together with Δm_{iso}, and the conjecture $\varepsilon_G = \varepsilon_G^B \in \mathbb{Q}(\sqrt{5})$ is proved to 17 ppm. As a consequence, Newton's gravitational constant becomes a fully combinatorial prediction from the proton quintuplet, with Δm_{iso} as the sole irreducible external input.

The 47 ppm residual is consistent with a QED correction of order $2\alpha\,\Delta m_{\mathrm{iso}}/m_p$, lying beyond the current PDL–QCD boundary and identified as a natural target for future work.

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